INDEPENDENT AUDIT AND ASSESSMENT · CLASSIFIED TURNING MOVEMENT SURVEY

New Sanganer Road, Jaipur

Six junctions, TMC-01 to TMC-06, between Mansarovar Metro and Sanganer Stadium. Surveyed 2026-05-11 and 2026-05-12 by the appointed contractor and issued to JDA as twelve workbooks. This is an independent re-derivation of every number in them, checked against the survey drawing. Road name and alignment are named in JDA's supplied KML. Prepared 26 August 2026.

253,440	225	+14.9%	12/12	1
CLASS-BINS RE-DERIVED	STORED TOTALS THAT DISAGREE	PCU UNDERSTATEMENT, FLOOR	U-TURN BAYS THAT FAIL	USABLE SURVEY DAY

1 What the issued survey gets wrong

Five findings, each re-derived from the workbooks rather than asserted. Every stored total was recomputed from its own components; where the two disagree the discrepancy is registered, never silently corrected.

FINDING	EVIDENCE	CONSEQUENCE
The second day was not independently observed	396 of 555 movement-by-class series reproduce day one to the exact vehicle. Of those that move, 154 rise and 5 fall; $p \approx 2\times10^{-39}$.	Treat as a one-day survey. Day-over-day growth measures the derivation.
The scheme's key movement was never counted	No U-turn column exists anywhere in the twelve workbooks. Twelve movements per junction, not sixteen.	The signal-free scheme converts right turns into U-turns and has no traffic evidence base for the movement it depends on.
PCU conversion understates demand	Two-wheelers are 48-53% of the stream and are carried at PCU 0.50, the value IRC:106 reserves for a class below 5%. The correct factor at that share is 0.75.	Corridor demand understated by at least 14.9%, band 13.7% to 74.8%.
Arithmetic does not close	225 stored totals disagree with their own components: 180 understate, 43 overstate.	Conservation between movement and approach sheets breaks. All registered.
The flow-diagram sheet is mislabelled	960 broken references across 12 files; the two-wheeler count is reported under the label Taxi.	Anyone reading the summary sheet rather than the movement sheets gets the wrong class split.

Survey design, separately: 11 May 2026 is a Monday; the methodology specifies Tue/Wed/Thu and excludes Monday. May is pre-monsoon peak heat; recommended windows are Oct-Nov or Feb-Mar. No weekend day surveyed; the stated minimum is three days including Sat and Sun.

2 Does the signal-free U-turn scheme work?

No, and not marginally. Under signal-free operation a right turn becomes a U-turn: the driver goes through, turns at a median bay, comes back and turns left. Each bay is fed by three movements, not one. Capacity of an unsignalised U-turn is gap acceptance, with the critical gap weighted by observed composition because two-wheelers accept far shorter gaps. **12 of 12 bays cannot be served**, and 12 sit past the point where acceptable gaps effectively cease to exist. The conclusion holds on all 12 published gap bases tested.

The finding a verdict alone hides

As the opposing flow falls to zero, gap-acceptance capacity tends to 3600 divided by the follow-up headway. A single median opening therefore passes at most **1,800 veh/hour with nothing at all to yield to**. **2 of the 12 bays are above that ceiling** - TMC-01 southbound at 2,755 and TMC-04 southbound at 1,889 veh/hour. Those are not badly sited bays. They are the wrong instrument for the demand, and no metering, median widening or opposing-flow relief reaches them.

The decision ladder: which constraint binds, bay by bay

Five criteria evaluated in order, first failure binding. A criterion below the binding one is reported untested, never passed - the order exists so that geometry is not checked for a problem geometry cannot reach.

CRITERION	BAYS IT BINDS AT	STATUS
gap capacity	12 of 12	binds first at every bay
median width	0 of 12	not reached
storage	0 of 12	not reached
weaving	0 of 12	not reached
detour burden	0 of 12	not reached

What can be done instead. Of the 5 measures tested, 3 can move the binding constraint here (keep the signal; signalise the U-turn; grade separate the through movement) and 2 cannot (relocate the bay; widen the median, add an acceleration lane). Grade separation is the only one that removes the conflicting flow rather than asking the U-turn to find gaps in it - and for the two bays above the ceiling it works by removing the need for a U-turn, not by making the bay work.

There is almost nowhere on this corridor to turn round

A right turn becomes four manoeuvres: past the junction, out to a median opening, through 180 degrees, back, then the left turn. But **5 of 6 median openings sit within 100 m of a junction centre** - they are junction mouths, not mid-block bays. Turning at one is not a detour; it is the driver turning AT the junction, which is the movement the scheme exists to remove. Only **1** genuine mid-block opening exists on 10 measurable bays' worth of corridor, so all seven proposed bays would have to be built new. The realistic detour is junction to junction: mean **1,647 m**, 2,605 extra vehicle-km in the peak hour, against a full range of 5 to 1,724 m. These remain the SHORTEST detours physically available, so every figure is a lower bound.

3 Capacity, and the one number that decides it

0/12	12/12	2047	55.8%
APPROACHES OVER CAPACITY TODAY	SURVIVE THE 2046 HORIZON	FIRST APPROACH BACK OVER	TRANSECTS READING SERVICE-ROAD WIDE

On the corrected alignment and the corrected widths this corridor is **not over capacity today**, and relief would hold to the 2046 horizon with the first approach returning to capacity in 2047. That is a weaker congestion case than an earlier version of this work stated, and it is stated plainly because it moved when the inputs were corrected.

It rests on one uncertain number. A transect cannot distinguish a through lane from a service road. 43 of 77 transects (56%) measure over 14 m per direction, up to 19.9 m, which is five running lanes each way. That is possible on this corridor and it is also what a service road looks like from above. Capacity scales linearly with this, so treat the wide junctions as an upper bound until the carriageway is confirmed on site.

4 How precise is any of this?

The supplied DWG contains no dimension entities. Nothing in it states a width. Every width, offset and chainage is scaled off georeferenced linework, and a scaled number printed to one decimal place is indistinguishable from a measured one. Each is therefore published with its method, its repeatability, and the field measurement that would settle it - 7 of 7 carry a stated uncertainty.

What testing the method against itself found

The transect spacing is an arbitrary choice inside our own method, so an answer that moves with it is measuring the choice rather than the road. At 25 m the whole corridor yields only 18 usable transects. **TMC-01 read 11.7 m off two of them and 15.6 m at every finer spacing** - a lane. Spacing is now 5 m, inside the converged region for all six. This was a defect in our work, found by testing it, and the published widths are the corrected ones.

JUNCTION	25 m	10 m	5 m	2 m	SETTLES AT
TMC-01	11.7	15.6	15.6	15.7	10 m
TMC-02	12.7	12.4	12.5	12.5	25 m
TMC-03	13.0	12.8	13.0	13.0	25 m
TMC-04	18.6	17.9	17.6	17.6	10 m
TMC-05	18.3	18.1	18.2	18.2	25 m
TMC-06	18.0	19.5	19.3	19.3	10 m

The two JDA sources agree. Distance from JDA's KML centreline to the nearest divider line in JDA's CAD: median 1.36 m, 90th percentile 3.92 m over 179 stations. Two independently produced descriptions of one corridor, consistent to about a metre. That is the evidence that the chainage and the transects share a geometry.

5 What automated analysis adds, and what it does not

Three applications, each shown with the test it could have failed. A model reported without one is a number with a confident voice.

APPLICATION	THE TEST IT HAD TO PASS	RESULT
Survey integrity screen	Six independent detectors must re-find the defects the audit proved by hand, without being told they exist.	PASSED 2 of 2. Found the duplicated second day (100 series identical in every live bin) and the 225 broken totals. Reusable on any future contractor submission.
Approach typology	The clusters must recover a label held out of the fitting: corridor arm versus cross-street arm. Two feature sets fitted, both published.	MIXED, honestly. Temporal shape found nothing (silhouette 0.21) - 24 approaches on one clock, which is what makes a single corridor-wide peak hour defensible. Composition separated cleanly (silhouette 0.559, held-out label recovered at 83% against 66% chance, p=0.0118).
How short a count can be	Beat doing no modelling at all, leave-one-out so no approach predicts itself.	PASSED 2 of 2. A 4-hour count (10:00-14:00) predicts the 24-hour total to 7.82% against a 52.55% no-model baseline. Worst single approach 22.12%. The next survey costs a fraction of this one, and JDA can spot-check a 24-hour submission against four hours of its own.

Limits, stated rather than buried: the count model is fitted on ONE independent day of ONE corridor, it forecasts nothing about a future year, and its window was chosen on the same error it is reported with over 18 combinations. The automated video counting stage is built and self-tested but has never seen real footage, so no accuracy figure is claimed for it.

6 Corrections we made to our own work

Listed because a reviewer is entitled to know which conclusions moved and why. Each was found by checking, not by being told.

WHAT WAS WRONG	HOW IT WAS FOUND	WHAT IT CHANGED
We had the wrong road. Six junctions were placed by matching arm names to signal clusters in the CAD.	JDA's reviewer challenged it, then supplied a KML.	Our picks sat 269 to 950 m off, on a parallel road. Rebuilding on JDA's alignment reversed most of the congestion case.
The U-turn scope counted right turns only.	JDA's reviewer pointed out a bay is fed by more than one movement.	Corrected to the full median U-turn: corridor right, cross-street right and cross-street through. Demand rose from 4,523 to 15,536 veh/hour.
Transect spacing of 25 m was too coarse to measure a width.	Re-running our own method across spacings.	TMC-01 went 11.7 m to 15.6 m - three lanes to four - and with it the capacity, v/c and design-life results in section 3.

WHAT WAS WRONG	HOW IT WAS FOUND	WHAT IT CHANGED
We argued a lane model does not fit because flow exceeds saturation flow.	The width correction above.	On corrected widths every approach runs below saturation. The argument is WITHDRAWN, not restated. The conclusion survives only on composition, which is weaker evidence, and the report says so.

7 Questions for JDA (12)

Every item here is something this analysis cannot settle from the data held. Each carries what blocks it and the one artefact that would unblock it. Things we simply have not built yet are deliberately not on this list.

#	QUESTION	WHY WE CANNOT SETTLE IT	WHAT WOULD SETTLE IT
1	**U-turn bay locations** What is the chainage of each of the seven proposed U-turn bays along the corridor alignment?	Three of the twelve bays we assess fall beyond the extent of the supplied drawing, so their detour is not measured at all. The rest are placed at the nearest median opening the CAD shows, which is the SHORTEST detour physically available, not the proposed one. Every detour figure we publish is therefore a lower bound.	A chainage or a coordinate for each of the seven bays.
2	**Chainage convention** Does JDA chain this corridor from the Sanganer Stadium end or the Mansarovar Metro end?	Our chainage runs from the north, which makes it count DOWN as the junction number counts up. That is the reverse of the survey's own numbering and it is confusing to read against JDA drawings.	One sentence, or any JDA drawing with a chainage marked on it.
3	**Service roads** Do the northern junctions carry service roads alongside the main carriageway?	4 of six junctions measure over 14 m per direction and 55.8% of transects do. That is five running lanes each way, or it is a service road being read as carriageway. Capacity scales linearly with the answer, and it is the single figure that decides whether this corridor is over capacity today.	Confirmation, or a typical cross-section for the northern half.
4	**Carriageway dimensions** Is there a dimensioned drawing or a survey control sheet for this corridor?	The supplied DWG carries no dimension entities. Every width, offset and chainage we publish is scaled from georeferenced linework and is provisional. We have quantified how repeatable that scaling is, which is not the same as knowing it is right.	A total station traverse with cross-sections at each junction and at each median opening, tied to stated control.
5	**The composite count columns** Can the survey's three composite columns be re-issued split into their IRC:106 classes?	Column B lumps car, auto-rickshaw and pickup at one PCU factor; two more columns do the same. That is why our PCU correction is a band (13.7% to 74.8%) rather than a number. The 14.9% floor is only the part that maps one to one.	The contractor's raw tally sheets, or a re-count to IRC:106 classes.
6	**Column heading, IRC:SP:41 Table 3.1** Should the column read "Three Wheeler (Auto), 3 Axle Truck, Buses"?	As issued it reads "Three Wheeler (Auto) Axle Truck, Buses" with no comma and no 3. If that is a typing error the column is the standard Table 3.1 row; if it is not, we do not know what was counted in it.	Confirmation of the intended heading.
7	**Pedestrians** Was a pedestrian count taken separately, and can it be issued?	IRC:SP:41 Table 3.1 carries a PEDESTRIAN Nos. row and clause 3.1(iv) requires it in urban areas with substantial pedestrian movement. The row is empty in all twelve workbooks. Removing signals removes the only protected crossing opportunity a pedestrian currently has.	The pedestrian sheets, or confirmation that none were taken.
8	**E-rickshaws** Under which column were e-rickshaws recorded?	No column carries them, though the label appears in the workbook's string table. If they were tallied into the car column their PCU is understated; if they were not counted the total is understated.	Confirmation from the enumerator instructions.

#	QUESTION	WHY WE CANNOT SETTLE IT	WHAT WOULD SETTLE IT
9	**The second survey day** Can the 12 May field sheets be produced?	396 of 555 movement-by-class series reproduce 11 May to the exact vehicle, and of those that move, 154 rise against 5 that fall. Under independent counting that split has a probability of about 2 in 10^39 - one in a billion billion billion billion. We treat this as a one-day survey.	The original tally sheets for 12 May.
10	**The flyover in the scheme video** Does the grade-separated structure shown in JDA's animation form part of this scheme, and over which junctions?	Our assessment tests two futures: the signal-free U-turn scheme as published, and an elevated through-carriageway. Which one is being designed changes which of our results is the relevant one.	The scheme drawing or a scope note.
11	**Critical gap** Is there any Indian measurement of U-turn critical gap on an urban arterial that JDA would accept?	Indo-HCM 2017 has no chapter, table or parameter set for mid-block median openings or U-turns; Annexure 7C is scoped to roundabouts. No Indian code publishes a design gap for the manoeuvre this scheme is built on. We used a composition-weighted gap and tested the conclusion across 12 separate published bases; it fails on all of them. But the parameter is the one an objector would attack first.	A cited value, or agreement to measure it on site.
12	**Bay geometry** What deceleration and storage length is proposed at each U-turn bay?	Gap capacity binds at all 12 bays, so storage is never reached in our ladder. If the gap problem were treated, storage would be the next criterion and we have no geometry for it.	The bay typical drawing.

8 What is still open at our end

Separate from the questions above, because these are ours to finish, not JDA's to answer.

- **Automated counting is unverified.** Detection, tracking, homography, zone counting and the two-stage training chain are built and pass their own gates against synthetic data. None has seen real footage, so the validation report stands as a pro forma with its gates published ahead of the measurement. Needs TMC-04 footage, ground-control stills and about 500 annotated frames.
- **Half the PCU correction is unresolvable** from the issued class scheme. Only a re-count to IRC:106 classes closes it - see question 5.
- **The corridor ordering is not settled by the counts alone.** Flow continuity ranks its best ordering by a 1.2% margin, which is noise. Chainage along JDA's alignment resolves it, and the agreement between the two is a check rather than a derivation.
- **The economic case for intervention is currently zero, and that is a finding, not a gap.** No approach is over capacity on the corrected widths, so there are no excess arrivals for the delay model to accumulate and the annual cost of delay comes out at nil - against 21 years before the first approach returns to capacity. A benefit-cost case for grade separation cannot be built on these numbers. It can be built on the narrower widths, which is why question 3 decides more than any other item on the list: at 55.8% of transects reading service-road wide, whether those are running lanes is what separates a corridor that needs a structure from one that does not.

Independent re-derivation from the twelve issued workbooks. 253,440 class-bins parsed; every stored total recomputed from components; discrepancies recorded rather than corrected. Every figure in this document is generated from the pipeline output, not transcribed. Standards referenced: IRC:106-1990, IRC:SP:41-1994, IRC:92-2017, Indo-HCM 2017. Analysis day 2026-05-11. Widths and chainages provisional pending a total station survey.